3. Tiga himpunan pelanggan

$A = \{N, N+2, N+4, N+7, N+10, N+13, N+15\} \rightarrow$ medsos

$B = \{N+3, N+5, N+7, N+10, N+12, N+18\} \rightarrow$ referral

$C = \{N+1, N+7, N+10, N+11, N+17, N+20\} \rightarrow$ loyalty program

$\Rightarrow = A = \{9, 11, 13, 16, 19, 22, 24\}$

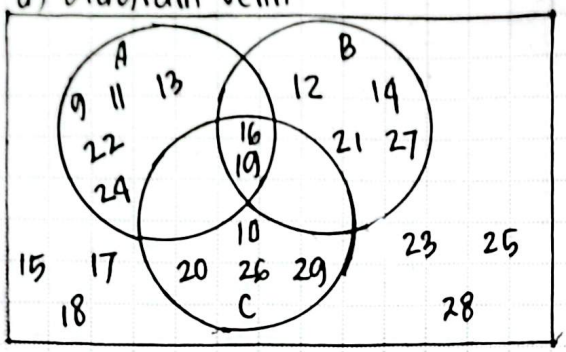
$B = \{12, 14, 16, 19, 21, 27\}$

$C = \{10, 16, 19, 20, 26, 29\}$

$U = \{9, 10, 11, 12, 13, 14, 15, 16, 17, 18, 19, 20, 21, 22, 23, 24, 25, 26, 27, 28, 29\}$

Dipindai dengan CamScanner

a) Diagram venn



- $|A \cup B \cup C| = \{9, 10, 11, 12, 13, 14, 16, 19, 20, 21, 22, 24, 26, 27, 29\} = 15$
- $|A \cap B \cap C| = \{16, 19\} = 2$
- $|(A \cup B) \cap C'| = \{9, 11, 12, 13, 14, 21, 22, 24, 27\} = 9$
- $|A \oplus B| = \{9, 11, 12, 13, 14, 21, 22, 24, 27\} = 9$

$C' = U - C = \{9, 11, 12, 13, 14, 15, 16, 17, 18, 19, 21, 22, 23, 24, 25, 27, 28\}$

$A \cup B = \{9, 11, 12, 13, 14, 16, 19, 21, 22, 24, 27\}$

$A \cap B = \{16, 19\}$

c) $f(x) = Kx + (a+b)$; $a = 0$

$g(x) = \dfrac{(x^2)}{(b+1)+a}$ $\quad b = 9$

$K = 10$

$\Rightarrow f(x) = 10x + 9$

$g(x) = \dfrac{x^2}{10}$

$\Rightarrow (f \circ g)(x) = f(g(x))$

$= \cancel{10}\left(\dfrac{x^2}{\cancel{10}}\right) + 9$

$= x^2 + 9$

$(g \circ f)(x) = g(f(x))$

$= \dfrac{(10x+9)^2}{10}$

$= \dfrac{100x^2 + 180x + 81}{10}$

$\Rightarrow x = b + 2$

$= 9 + 2$

$x = 11$

$f(g(x)) = 11^2 + 9$

$= 121 + 9$

$= 130$

$g(f(x)) = \dfrac{(10(11) + 9)^2}{10}$

$= \dfrac{(119)^2}{10}$

$= 1416,1$

tidak komutatif

Dipindai dengan CamScanner

3. d) $f^{-1}(x) \rightarrow \frac{x-9}{10}$

$\Rightarrow y = 10x + 9$

$y - 9 = 10x$

$\frac{y-9}{10} = x$

$\Rightarrow$ Rp $(k \cdot 100 + a + b)$ ribu

$= 10 \cdot 100 + 9$

Rp 1009 ribu $\approx$ 1.009.000

$\Rightarrow x = f'(1.009.000)$

$= \frac{1.009.000 - 9}{10}$

$= \frac{1.008.991}{10}$

$= 1.008.99,1 \approx 1.009.000$ //

e) $f(x) = 10x + 9$

- Injektif karena $10x_1 + 9 = 10x_2 + 9$, maka $x_1 = x_2$
- Surjektif karena setiap $y \in \mathbb{R}$, bisa menggunakan $x = \frac{y-9}{10}$, maka $f(x) = y$
- Bijektif karena bisa injektif dan surjektif

$g(x) = \frac{x^2}{10}$

- tidak injektif, $g(1) = 0,1$ dan $g(-1) = 0,1$, $1 \neq -1$ tapi menghasilkan hasil yang sama
- tidak surjektif, $x = -1 \in \mathbb{R}$, tidak ada x real dengan $x^2 = -1$ $(\frac{x^2}{10} \geq 0)$
- tidak bijektif karena tidak injektif dan tidak surjektif

Dipindai dengan CamScanner